Transformations and congruence

ANSWER KEY

Core Concepts: Transformations and congruence

1 In $\triangle ABC$ studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 103° ✓

b. 113°

c. 95°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 45^\circ) = 103^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 8

c. 6 ✓

d. 9

Substituting $x = 1$ into $y = 3x + 3$ gives $y = 3(1) + 3 = 6$.

Procedural Fluency & Algebraic Manipulation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 In $\triangle ABC$ studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 96° ✓

b. 106°

c. 88°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 52^\circ) = 96^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 6

b. 11

c. 9 ✓

d. 12

Substituting $x = 2$ into $y = 3x + 3$ gives $y = 3(2) + 3 = 9$.

Graphical & Geometric Interpretation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 In $\triangle ABC$ studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 89° ✓

b. 99°

c. 81°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 59^\circ) = 89^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm ✓

c. 30 cm

d. 96 cm

Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 9

b. 14

c. 12 ✓

d. 15

Substituting $x = 3$ into $y = 3x + 3$ gives $y = 3(3) + 3 = 12$.

Exam-Style Problem Solving & Modeling

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 In $\triangle ABC$ studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 82° ✓

b. 92°

c. 74°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 66^\circ) = 82^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm ✓

c. 36 cm

d. 150 cm

Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 12

b. 17

c. 15 ✓

d. 18

Substituting $x = 4$ into $y = 3x + 3$ gives $y = 3(4) + 3 = 15$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647